7:11 AM

Quiz
vulms.vu.edu.pk

Question # 1 of 10 ( Start time: 07:11:35 AM, 30 May 2023 )

The time taken by the particle to complete one oscillation is called ------.

Select the correct option

- frequency
- time period
- amplitude
- none of these

Click to Sav

7:12 AM

846 B/s

Quiz
vulms.vu.edu.pk

BC210202095: MUQADAS BATOOL

Time Left 54 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 07:11 AM, 30 May 2023

Question # 2 of 10 ( Start time: 07:12:02 AM, 30 May 2023 )

Total Marks: 1

frequency is reciprocal of -----.

Select the correct option

- wavelength
- time period
- cycle
- amplitude

Click to Save Answer & Move to Next Question

7:12 AM

157 B/s

76

Quiz
vulms.vu.edu.pk

BC210202095: MUQADAS BATOOL

Time Left 79 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 07:11 AM, 30 May 2023

Question # 3 of 10 ( Start time: 07:12:42 AM, 30 May 2023 )

Total Marks: 1

When the acceleration is a function of distance only, we may write

$$a(x) = ----$$

Select the correct option

Reload Math Equations

- None of these
- $\frac{d^2x}{dt^2}$
- $\frac{dx}{dt}$
- $\frac{dv}{dt}$

Click to Save Answer & Move to Next Question

7:13 AM

174 B/s

Quiz
vulms.vu.edu.pk

BC210202095: MUQADAS BATOOL

Time Left 50 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 07:11 AM, 30 May 2023

**Question # 4 of 10 ( Start time: 07:13:03 AM, 30 May 2023 )** **Total Marks: 1**

A particle is said to execute simple harmonic motion if it moves in a straight line so that its acceleration is always directed towards a fixed point in the line and is proportional to the ----- of the particle from the fixed point.

**Select the correct option**

- none of these
- square root of the distance
- square of the distance
- distance

Click to Save Answer & Move to Next Question

7:13 AM
15.5 K/s
75
Quiz
vulms.vu.edu.pk

**Question # 5 of 10 ( Start time: 07:13:46 AM, 30 May 2023 )**

The time taken by the particle to complete one oscillation is called -----.

**Select the correct option**

- amplitude
- time period
- frequency
- none of these

Click to Save A

7:14 AM

Quiz
vulms.vu.edu.pk

BC210202095: MUQADAS BATOOL

Time Left 68 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 07:11 AM, 30 May 2023

Question # 6 of 10 ( Start time: 07:14:04 AM, 30 May 2023 )

Total Marks: 1

The space-time curve is a ------ for a constant velocity.

Select the correct option

- circle
- hyperbola
- straight line
- parabola

Click to Save Answer & Move to Next Question

7:14 AM

Quiz
vulms.vu.edu.pk

BC210202095: MUQADAS BATOOL

Time Left 80 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 07:11 AM, 30 May 2023

Question # 7 of 10 ( Start time: 07:14:30 AM, 30 May 2023 )

Total Marks: 1

At mean position of pendulum, the potential energy of the pendulum is:

Select the correct option

- maximum
- minimum
- neither maximum nor minimum
- none of these

Click to Save Answer & Move to Next Question

7:15 AM

Quiz
vulms.vu.edu.pk

BC210202095: MUQADAS BATOOL

Time Left 34 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 07:11 AM, 30 May 2023

Question # 8 of 10 ( Start time: 07:14:44 AM, 30 May 2023 )

Total Marks: 1

If the displacement of a particle is given by

$$x = 3t^2 + 4$$

, then the acceleration of the particle at any time t will be -------.

Select the correct option

Reload Math Equations

- 6t
- 3t
- 0
- 6

Click to Save Answer & Move to Next Question

7:15 AM

17.1 K/s

Quiz
vulms.vu.edu.pk

BC210202095: MUQADAS BATOOL

Time Left 82 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 07:11 AM, 30 May 2023

Question # 9 of 10 ( Start time: 07:15:45 AM, 30 May 2023 )

Total Marks: 1

A body of mass 5 g is executing SHM about a point O with amplitude 100 cm. If its max. velocity is 100 cm/sec, its velocity will be 50 cm/sec at a distance (in cm):

Select the correct option

Reload Math Equations

- $50\sqrt{2}$
- $50\sqrt{3}$
- $10\sqrt{3}$
- 5

Click to Save Answer & Move to Next Question

7:16 AM

Quiz
vulms.vu.edu.pk

BC210202095: MUQADAS BATOOL

Time Left 41 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 07:11 AM, 30 May 2023

Question # 10 of 10 ( Start time: 07:15:57 AM, 30 May 2023 )

Total Marks: 1

If the acceleration is 5m/s2 throughout the motion, the velocity-time curve will be a -----

Select the correct option

- circle
- straight line
- parabola
- hyperbola

Click to Save Answer & Move to Next Question

8:41 AM

Quiz
vulms.vu.edu.pk

BC210206267: USMAN MUNIR MAHMOOD

Time Left 64 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:41 AM, 30 May 2023

**Question #1 of 10 ( Start time: 08:41:22 AM, 30 May 2023 )** **Total Marks: 1**

If the velocity of the particle remains same throughout the motion, the slope of the space-time curve -----.

**Select the correct option**

- Does not remain the same
- varies
- Remains the same
- None of these

Click to Save Answer & Move to Next Question

8:42 AM

Quiz
vulms.vu.edu.pk

BC210206267: USMAN MUNIR MAHMOOD

Time Left 67 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:41 AM, 30 May 2023

Question # 2 of 10 ( Start time: 08:41:52 AM, 30 May 2023 )

Total Marks: 1

The maximum distance covered by the particle executing simple harmonic motion in either side of mean position is called as ------.

Select the correct option

- none of these
- time period
- amplitude
- frequency

Click to Save Answer & Move to Next Question

8:43 AM

156 B/s

88

Quiz
vulms.vu.edu.pk

BC210206267: USMAN MUNIR MAHMOOD

Time Left 32 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:41 AM, 30 May 2023

Question # 3 of 10 ( Start time: 08:42:19 AM, 30 May 2023 )

Total Marks: 1

The space-time curve is a ------ for a constant velocity.

Select the correct option

straight line

parabola

hyperbola

circle

Click to Save Answer & Move to Next Question

8:44 AM

Quiz
vulms.vu.edu.pk

BC210206267: USMAN MUNIR MAHMOOD

Time Left 63 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:41 AM, 30 May 2023

Question # 5 of 10 ( Start time: 08:43:56 AM, 30 May 2023 )

Total Marks: 1

At mean position of pendulum, the potential energy of the pendulum is:

Select the correct option

- none of these
- neither maximum nor minimum
- maximum
- minimum

Click to Save Answer & Move to Next Question

8:44 AM

15.5 K/s

Quiz
vulms.vu.edu.pk

BC210206267: USMAN MUNIR MAHMOOD

Time Left 85 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:41 AM, 30 May 2023

**Question # 6 of 10 ( Start time: 08:44:27 AM, 30 May 2023 )** **Total Marks: 1**

If the acceleration is 5m/s2 throughout the motion, the velocity-time curve will be a -----.

**Select the correct option**

- parabola
- hyperbola
- straight line
- circle

Click to Save Answer & Move to Next Question

8:44 AM

Quiz
vulms.vu.edu.pk

BC210206267: USMAN MUNIR MAHMOOD

Time Left 85 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:41 AM, 30 May 2023

Question # 7 of 10 ( Start time: 08:44:36 AM, 30 May 2023 )

Total Marks: 1

If the displacement of a particle is given by

$$x = 3t^2 + 4$$

, then the acceleration of the particle at any time t will be -------.

Select the correct option

Reload Math Equations

- 6
- 0
- 3t
- 6t

Click to Save Answer & Move to Next Question

8:45 AM

Quiz
vulms.vu.edu.pk

BC210206267: USMAN MUNIR MAHMOOD

Time Left 80 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:41 AM, 30 May 2023

Question # 8 of 10 ( Start time: 08:44:56 AM, 30 May 2023 )

Total Marks: 1

frequency is reciprocal of -----.

Select the correct option

- wavelength
- amplitude
- time period
- cycle

Click to Save Answer & Move to Next Question

8:45 AM

Quiz
vulms.vu.edu.pk

BC210206267: USMAN MUNIR MAHMOOD

Time Left 64 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:41 AM, 30 May 2023

**Question # 9 of 10 ( Start time: 08:45:10 AM, 30 May 2023 )** **Total Marks: 1**

A particle is said to execute simple harmonic motion if it moves in a straight line so that its acceleration is always directed towards a fixed point in the line and is proportional to the ----- of the particle from the fixed point.

**Select the correct option**

- square of the distance
- none of these
- square root of the distance
- distance

Click to Save Answer & Move to Next Question

8:45 AM

821 B/s

Quiz
vulms.vu.edu.pk

BC210206267: USMAN MUNIR MAHMOOD

Time Left 76 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:41 AM, 30 May 2023

Question # 10 of 10 ( Start time: 08:45:40 AM, 30 May 2023 )

Total Marks: 1

The time taken by the particle to complete one oscillation is called ------.

Select the correct option

- frequency
- none of these
- amplitude
- time period

Click to Save Answer & Move to Next Question

8:48 AM

Quiz
vulms.vu.edu.pk

BC210201988: MUHAMMAD AHSAN NAWAZ

MTH404 - Dynamics (Quiz #1) Quiz St

**Question # 1 of 10 ( Start time: 08:47:18 AM, 30 May 2023 )**

During S.H.M kinetic energy of the body is ----- at mean position.

**Select the correct option**

- maximum
- zero
- none of these
- minimum

Click to Save An

8:48 AM

Quiz
vulms.vu.edu.pk

**Question # 2 of 10 ( Start time: 08:48:05 AM, 30 May 2023 )**

When the acceleration is a function of distance only, we may write

$$a(x) = - - - -$$

**Select the correct option**

| | |
|---|---|
| ○ | $\frac{dx}{dt}$ |
| ● | $\frac{d^2x}{dt^2}$ |
| ○ | $\frac{dv}{dt}$ |
| ○ | None of these |

Click to Save Answ

8:48 AM

Quiz
vulms.vu.edu.pk

BC210201988: MUHAMMAD AHSAN NAWAZ

Time Left 79 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:47 AM, 30 May 2023

Question # 3 of 10 ( Start time: 08:48:30 AM, 30 May 2023 )

Total Marks: 1

Time period is reciprocal of-----.

Select the correct option

- (•) frequency
- ( ) amplitude
- ( ) cycle
- ( ) wavelength

Click to Save Answer & Move to Next Question

8:48 AM

13.3 K/s

Quiz
vulms.vu.edu.pk

BC210201988: MUHAMMAD AHSAN NAWAZ

Time Left 86 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:47 AM, 30 May 2023

Question # 4 of 10 ( Start time: 08:48:44 AM, 30 May 2023 )

Total Marks: 1

If the displacement of a particle is given by

$$x = 3t^2 + 4$$

, then the acceleration of the particle at any time t will be -------.

Select the correct option

Reload Math Equations

- 0
- 6t
- 3t
- 6

Click to Save Answer & Move to Next Question

8:48 AM

13.2 K/s

88

Quiz
vulms.vu.edu.pk

BC210201988: MUHAMMAD AHSAN NAWAZ

Time Left 87 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:47 AM, 30 May 2023

Question # 5 of 10 ( Start time: 08:48:53 AM, 30 May 2023 )

Total Marks: 1

frequency is reciprocal of -----.

Select the correct option

- wavelength
- time period
- cycle
- amplitude

Click to Save Answer & Move to Next Question

8:50 AM

Quiz
vulms.vu.edu.pk

BC210201988: MUHAMMAD AHSAN NAWAZ

Time Left 79 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:47 AM, 30 May 2023

Question # 7 of 10 ( Start time: 08:50:16 AM, 30 May 2023 )

Total Marks: 1

The time taken by the particle to complete one oscillation is called -----.

Select the correct option

- none of these
- frequency
- time period
- amplitude

Click to Save Answer & Move to Next Question

8:50 AM

Quiz
vulms.vu.edu.pk

BC210201988: MUHAMMAD AHSAN NAWAZ

Time Left 85 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:47 AM, 30 May 2023

Question # 8 of 10 ( Start time: 08:50:32 AM, 30 May 2023 )

Total Marks: 1

If the acceleration is 5m/s2 throughout the motion, the velocity-time curve will be a -----.

Select the correct option

- straight line
- hyperbola
- circle
- parabola

Click to Save Answer & Move to Next Question

8:51 AM

Quiz
vulms.vu.edu.pk

BC210201988: MUHAMMAD AHSAN NAWAZ

Time Left 73 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:47 AM, 30 May 2023

Question # 9 of 10 ( Start time: 08:50:51 AM, 30 May 2023 )

Total Marks: 1

The maximum distance covered by the particle executing simple harmonic motion in either side of mean position is called as ------.

Select the correct option

- time period
- amplitude
- frequency
- none of these

Click to Save Answer & Move to Next Question

8:51 AM

Quiz
vulms.vu.edu.pk

BC210201988: MUHAMMAD AHSAN NAWAZ

Time Left 86 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:47 AM, 30 May 2023

Question # 10 of 10 ( Start time: 08:51:12 AM, 30 May 2023 )

Total Marks: 1

A body of mass 5 g is executing SHM about a point O with amplitude 100 cm. If its max. velocity is 100 cm/sec, its velocity will be 50 cm/sec at a distance (in cm):

Select the correct option

Reload Math Equations

- 5
- $10\sqrt{3}$
- $50\sqrt{3}$
- $50\sqrt{2}$

Click to Save Answer & Move to Next Question

8:59 AM

Quiz
vulms.vu.edu.pk

BC210206353: AYMAN SHEHZAD

Time Left 84 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:58 AM, 30 May 2023

Question #1 of 10 ( Start time: 08:58:54 AM, 30 May 2023 )

Total Marks: 1

At mean position of pendulum, the potential energy of the pendulum is:

Select the correct option

- neither maximum nor minimum
- maximum
- minimum
- none of these

Click to Save Answer & Move to Next Question

[redacted]

8:59 AM

532 B/s

87

Quiz
vulms.vu.edu.pk

BC210206353: AYMAN SHEHZAD

Time Left 80 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:58 AM, 30 May 2023

Question # 2 of 10 ( Start time: 08:59:13 AM, 30 May 2023 )

Total Marks: 1

If the velocity of a particle is given by

$$v = 3t^2 + 4$$

, then the acceleration of the particle at time '3' seconds will be ------.

Select the correct option

Reload Math Equations

- 3
- 0
- 6
- 18

Click to Save Answer & Move to Next Question

8:59 AM

16.5 K/s

87

Quiz
vulms.vu.edu.pk

BC210206353: AYMAN SHEHZAD

Time Left 86 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:58 AM, 30 May 2023

Question # 3 of 10 ( Start time: 08:59:26 AM, 30 May 2023 )

Total Marks: 1

If the acceleration is 5m/s2 throughout the motion, the velocity-time curve will be a -----.

Select the correct option

- hyperbola
- straight line
- parabola
- circle

Click to Save Answer & Move to Next Question

8:59 AM

Quiz
vulms.vu.edu.pk

BC210206353: AYMAN SHEHZAD

Time Left 87 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:58 AM, 30 May 2023

Question # 4 of 10 ( Start time: 08:59:34 AM, 30 May 2023 )

Total Marks: 1

The space-time curve is a ------ for a constant velocity.

Select the correct option

- straight line
- circle
- hyperbola
- parabola

Click to Save Answer & Move to Next Question

8:59 AM

13.6 K/s

87

Quiz
vulms.vu.edu.pk

BC210206353: AYMAN SHEHZAD

Time Left 87 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:58 AM, 30 May 2023

Question # 5 of 10 ( Start time: 08:59:41 AM, 30 May 2023 )

Total Marks: 1

A body of mass 5 g is executing SHM about a point O with amplitude 100 cm. If its max. velocity is 100 cm/sec, its velocity will be 50 cm/sec at a distance (in cm):

Select the correct option

Reload Math Equations

- $50\sqrt{3}$
- 5
- $10\sqrt{3}$
- $50\sqrt{2}$

Click to Save Answer & Move to Next Question

9:00 AM

289 B/s 87

Quiz
vulms.vu.edu.pk

BC210206353: AYMAN SHEHZAD

Time Left 33 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:58 AM, 30 May 2023

Question # 6 of 10 ( Start time: 08:59:53 AM, 30 May 2023 )

Total Marks: 1

The maximum distance covered by the particle executing simple harmonic motion in either side of mean position is called as ------.

Select the correct option

- time period
- amplitude
- none of these
- frequency

Click to Save Answer & Move to Next Question

9:01 AM

Quiz
vulms.vu.edu.pk

BC210206353: AYMAN SHEHZAD

Time Left 81 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:58 AM, 30 May 2023

Question # 7 of 10 ( Start time: 09:01:08 AM, 30 May 2023 )

Total Marks: 1

Time period is reciprocal of-----.

Select the correct option

- wavelength
- frequency
- amplitude
- cycle

Click to Save Answer & Move to Next Question

9:01 AM

Quiz
vulms.vu.edu.pk

BC210206353: AYMAN SHEHZAD

Time Left 83 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:58 AM, 30 May 2023

Question # 8 of 10 ( Start time: 09:01:21 AM, 30 May 2023 )

Total Marks: 1

During S.H.M kinetic energy of the body is ----- at mean position.

Select the correct option

- minimum
- zero
- maximum
- none of these

Click to Save Answer & Move to Next Question

9:01 AM

13.2 K/s

Quiz
vulms.vu.edu.pk

BC210206353: AYMAN SHEHZAD

Time Left 87 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:58 AM, 30 May 2023

Question # 9 of 10 ( Start time: 09:01:37 AM, 30 May 2023 )

Total Marks: 1

When the acceleration is a function of distance only, we may write

$$a(x) = - - - -$$

Select the correct option

Reload Math Equations

- $\frac{d^2x}{dt^2}$
- $\frac{dv}{dt}$
- $\frac{dx}{dt}$
- None of these

Click to Save Answer & Move to Next Question

9:01 AM

Quiz
vulms.vu.edu.pk

BC210206353: AYMAN SHEHZAD

Time Left 75 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 08:58 AM, 30 May 2023

Question # 10 of 10 ( Start time: 09:01:44 AM, 30 May 2023 )

Total Marks: 1

The time taken by the particle to complete one oscillation is called -----.

Select the correct option

- none of these
- frequency
- amplitude
- time period

Click to Save Answer & Move to Next Question

9:28

vulms.vu.edu.pk/Quiz/Q

BC210207002: AMER GULZAR

Time Left 88 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 09:22 AM, 30 May 2023

Question # 7 of 10 ( Start time: 09:28:23 AM, 30 May 2023 )

Total Marks: 1

The time taken by the particle to complete one oscillation is called -----.

Select the correct option

- frequency
- none of these
- amplitude
- time period

Click to Save Answer & Move to Next Question

9:29

vulms.vu.edu.pk/Quiz/Q

BC210207002: AMER GULZAR

Time Left 88 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 09:22 AM, 30 May 2023

Question # 8 of 10 ( Start time: 09:29:26 AM, 30 May 2023 )

Total Marks: 1

At mean position of pendulum, the potential energy of the pendulum is:

Select the correct option

- neither maximum nor minimum
- none of these
- minimum
- maximum

Click to Save Answer & Move to Next Question

9:30

vulms.vu.edu.pk/Quiz/Q

BC210207002: AMER GULZAR

Time Left 88 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 09:22 AM, 30 May 2023

Question # 9 of 10 ( Start time: 09:30:13 AM, 30 May 2023 )

Total Marks: 1

If the velocity of a particle is given by

$$v = 3t^2 + 4$$

, then the acceleration of the particle at time '3' seconds will be ------.

Select the correct option

Reload Math Equations

- 3
- 0
- 6
- 18

Click to Save Answer & Move to Next Question

9:31

vulms.vu.edu.pk/Quiz/Q

BC210207002: AMER GULZAR

Time Left 88 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 09:22 AM, 30 May 2023

Question # 10 of 10 ( Start time: 09:31:01 AM, 30 May 2023 )

Total Marks: 1

If the velocity of the particle remains same throughout the motion, the slope of the space-time curve -----.

Select the correct option

- None of these
- Does not remain the same
- varies
- Remains the same

Click to Save Answer & Move to Next Question

9:23

vulms.vu.edu.pk/Quiz/Q

BC210207002: AMER GULZAR

Time Left 88 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 09:22 AM, 30 May 2023

Question # 2 of 10 ( Start time: 09:23:27 AM, 30 May 2023 )

Total Marks: 1

Time period is reciprocal of-----.

Select the correct option

- frequency
- cycle
- wavelength
- amplitude

Click to Save Answer & Move to Next Question

9:22

vulms.vu.edu.pk/Quiz/Q

BC210207002: AMER GULZAR

Time Left 87 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 09:22 AM, 30 May 2023

Question # 1 of 10 ( Start time: 09:22:31 AM, 30 May 2023 )

Total Marks: 1

frequency is reciprocal of -----.

Select the correct option

- cycle
- time period
- amplitude
- wavelength

Click to Save Answer & Move to Next Question

9:25

vulms.vu.edu.pk/Quiz/Q

BC210207002: AMER GULZAR

Time Left 88 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 09:22 AM, 30 May 2023

Question # 4 of 10 ( Start time: 09:25:14 AM, 30 May 2023 )

Total Marks: 1

During S.H.M kinetic energy of the body is ----- at mean position.

Select the correct option

- zero
- none of these
- maximum
- minimum

Click to Save Answer & Move to Next Question

9:24

vulms.vu.edu.pk/Quiz/Q

BC210207002: AMER GULZAR

Time Left 88 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 09:22 AM, 30 May 2023

Question # 3 of 10 ( Start time: 09:24:12 AM, 30 May 2023 )

Total Marks: 1

If the displacement of a particle is given by

$$x = 3t^2 + 4$$

, then the acceleration of the particle at any time t will be ------.

Select the correct option

Reload Math Equations

- 0
- 3t
- 6t
- 6

Click to Save Answer & Move to Next Question

9:26

vulms.vu.edu.pk/Quiz/Q

BC210207002: AMER GULZAR

Time Left 87 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 09:22 AM, 30 May 2023

Question # 5 of 10 ( Start time: 09:26:37 AM, 30 May 2023 )

Total Marks: 1

When the acceleration is a function of distance only, we may write

$$a(x) = - - - -$$

Select the correct option

Reload Math Equations

- $\frac{d^2x}{dt^2}$
- None of these
- $\frac{dv}{dt}$
- $\frac{dx}{dt}$

Click to Save Answer & Move to Next Question

9:27

vulms.vu.edu.pk/Quiz/Q

BC210207002: AMER GULZAR

Time Left 88 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 09:22 AM, 30 May 2023

Question # 6 of 10 ( Start time: 09:27:38 AM, 30 May 2023 )

Total Marks: 1

The space-time curve is a ------ for a constant velocity.

Select the correct option

- straight line
- circle
- parabola
- hyperbola

Click to Save Answer & Move to Next Question

9:27

vulms.vu.edu.pk/Quiz/Q

BC210207002: AMER GULZAR

Time Left 88 sec(s)

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 09:22 AM, 30 May 2023

Question # 6 of 10 ( Start time: 09:27:38 AM, 30 May 2023 )

Total Marks: 1

The space-time curve is a ------- for a constant velocity.

Select the correct option

- straight line
- circle
- parabola
- hyperbola

Click to Save Answer & Move to Next Question

10:10 82

vulms.vu.edu.pk/Quiz/(

3C210202256: SIDRA MANSHA

Time Left

MTH404 - Dynamics (Quiz #1)

Quiz Start Time: 10:10 AM, 30 May

Question # 1 of 10 ( Start time: 10:10:26 AM, 30 May 2023 )

Total M

A particle is said to execute simple harmonic motion if it moves in a straight line so that its acceleration is always directed towards a fixed p in the line and is proportional to the ----- of the particle from the fixed point.

Select the correct option

- none of these
- square root of the distance
- square of the distance
- distance

Click to Save Answer & Move to Next Questi

10:12 vulms.vu.edu.pk/Quiz/( 7

MTH404 - Dynamics (Quiz #1)

**Question # 2 of 10 ( Start time: 10:11:53 AM, 30 May 2023 )**

At mean position of pendulum, the potential energy of the pendulum is:

**Select the correct option**

- (●) minimum
- ( ) neither maximum nor minimum
- ( ) maximum
- ( ) none of these

Cl

10:06 AM
Quiz
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MTH404 - Dynamics (Quiz #1)

**Question # 10 of 10 ( Start time: 10:06:22 AM, 30 May 2023 )**

frequency is reciprocal of -----.

**Select the correct option**

- time period
- wavelength
- cycle
- amplitude

**Question # 9 of 10 ( Start time: 10:05:44 AM, 30 May 2023 )**

At mean position of pendulum, the potential energy of the pendulum is:

**Select the correct option**

- neither maximum nor minimum
- none of these
- maximum
- minimum

Click to Save Answer &

Question # 8 of 10 ( Start time: 10:04:52 AM, 30 May 2023 )

The maximum distance covered by the particle executing simple harmonic motion in either side of mean position is called as ----

Select the correct option

- amplitude
- time period
- frequency
- none of these

Click to Save Answer & Move to Ne

10:04 AM
Quiz
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Question # 7 of 10 ( Start time: 10:04:16 AM, 30 May 2023 )

The space-time curve is a ------ for a constant velocity.

Select the correct option

- straight line
- circle
- parabola
- hyperbola

Click to Save Answer &

MTH404 - Dynamics (Quiz #1) Quiz Start Time: 10:00 AM, 30 May 2023

Question # 6 of 10 ( Start time: 10:03:33 AM, 30 May 2023 ) Total Marks: 1

A body of mass 5 g is executing SHM about a point O with amplitude 100 cm. If its max. velocity is 100 cm/sec, its velocity will be 50 cm/sec at a distance (in cm):

Select the correct option

Reload Math Equations

- $10\sqrt{3}$
- $50\sqrt{3}$
- $50\sqrt{2}$
- 5

Click to Save Answer & Move to Next Question

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10:03 AM
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MTH404 - Dynamics (Quiz #1) Quiz Star

Question # 5 of 10 ( Start time: 10:03:03 AM, 30 May 2023 )

If the acceleration is 5m/s2 throughout the motion, the velocity-time curve will be a -----.

Select the correct option

- circle
- straight line
- hyperbola
- parabola

Click to Save Answ

10:02 AM
Quiz
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**Question # 4 of 10 ( Start time: 10:02:35 AM, 30 May 2023 )**

The time taken by the particle to complete one oscillation is called ------.

**Select the correct option**

- none of these
- time period
- frequency
- amplitude

Click to Save An

MTH404 - Dynamics (Quiz #1) Quiz Start Time: 10:0

Question # 3 of 10 ( Start time: 10:01:49 AM, 30 May 2023 )

The time taken by the particle to complete one oscillation is called -----.

Select the correct option

- frequency
- none of these
- time period
- amplitude

Click to Save Answer & Move t

10:00 AM
Quiz
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**Question # 2 of 10 ( Start time: 10:00:47 AM, 30 May 2023 )**

If the velocity of a particle is given by

$$v = 3t^2 + 4$$

, then the acceleration of the particle at time '3' seconds will be ------.

**Select the correct option**

| | |
|---|---|
| ○ | 3 |
| ○ | 6 |
| ○ | 0 |
| ○ | 18 |

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MTH404 - Dynamics (Quiz #1)

**Question # 1 of 10 ( Start time: 10:00:05 AM, 30 May 2023 )**

When the acceleration is a function of distance only, we may write

$$a(x) = - - - -$$

**Select the correct option**

| | |
|---|---|
| ○ | $\frac{dv}{dt}$ |
| ○ | $\frac{d^2x}{dt^2}$ |
| ○ | None of these |
| ○ | $\frac{dx}{dt}$ |

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MTH404 - Dynamics (Quiz #1) Quiz Start

Question # 10 of 10 ( Start time: 09:56:18 AM, 30 May 2023 )

If the velocity of a particle is given by

$$v = 3t^2 + 4$$

, then the acceleration of the particle at time '3' seconds will be ------.

Select the correct option

- 6
- 18
- 0
- 3

Click to Save Answ

9:55 vulms.vu.edu.pk/Quiz/( 7

MTH404 - Dynamics (Quiz #1)

Question # 9 of 10 ( Start time: 09:55:36 AM, 30 May 2023 )

During S.H.M kinetic energy of the body is ----- at mean position.

Select the correct option

- minimum
- none of these
- zero
- maximum

Click to

9:55 vulms.vu.edu.pk/Quiz/( 7

Question # 8 of 10 ( Start time: 09:55:06 AM, 30 May 2023 )

The time taken by the particle to complete one oscillation is called ------.

Select the correct option

- frequency
- amplitude
- none of these
- time period

Click to Save

9:54 vulms.vu.edu.pk/Quiz/( 7

MTH404 - Dynamics (Quiz #1)

Question # 7 of 10 ( Start time: 09:54:33 AM, 30 May 2023 )

The space-time curve is a ------ for a constant velocity.

Select the correct option

- circle
- hyperbola
- parabola
- straight line

Click

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**Question # 6 of 10 ( Start time: 09:53:58 AM, 30 May 2023 )**

If the displacement of a particle is given by

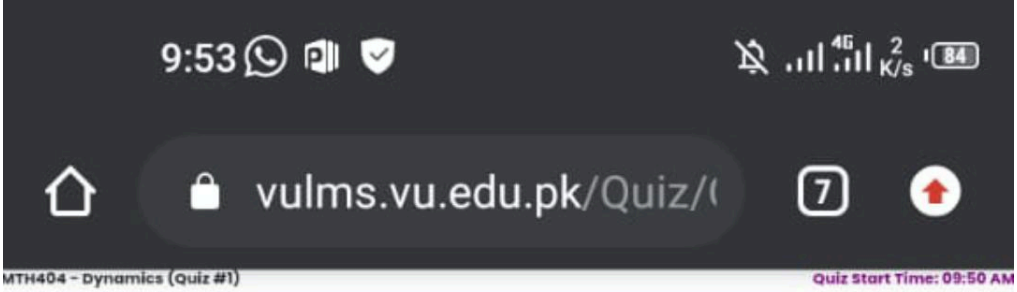
$$x = 3t^2 + 4$$

, then the acceleration of the particle at any time t will be ------.

**Select the correct option**

- 0
- 6t
- 3t
- 6

Click to Save An




MTH404 - Dynamics (Quiz #1) Quiz Start Time: 09:50 AM

Question # 5 of 10 ( Start time: 09:53:00 AM, 30 May 2023 )

A particle is said to execute simple harmonic motion if it moves in a straight line so that its acceleration is always directed toward in the line and is proportional to the ----- of the particle from the fixed point.

Select the correct option

- square of the distance
- distance
- square root of the distance
- none of these

Click to Save Answer & Move to Ne

9:52 vulms.vu.edu.pk/Quiz/( 7

Question # 4 of 10 ( Start time: 09:52:30 AM, 30 May 2023 )

Time period is reciprocal of-----.

Select the correct option

- amplitude
- wavelength
- cycle
- frequency

Click to Sa

9:51 vulms.vu.edu.pk/Quiz/( 7

MTH404 - Dynamics (Quiz #1)

Question # 3 of 10 ( Start time: 09:51:51 AM, 30 May 2023 )

If the acceleration is 5m/s2 throughout the motion, the velocity-time curve will be a -----.

Select the correct option

- parabola
- circle
- hyperbola
- straight line

9:51 149 B/s 85
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MTH404 - Dynamics (Quiz #1)

Question # 2 of 10 ( Start time: 09:51:25 AM, 30 May 2023 )

frequency is reciprocal of -----.

Select the correct option

- wavelength
- amplitude
- cycle
- time period

Cl

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7

MTH404 - Dynamics (Quiz #1) Quiz Start Time: 09:50

Question # 1 of 10 ( Start time: 09:50:31 AM, 30 May 2023 )

The maximum distance covered by the particle executing simple harmonic motion in either side of mean position is called as –

Select the correct option

- none of these
- frequency
- time period
- amplitude

Click to Save Answer & Move to